DEVANESH S M

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Profile Summary

AI and Data Science student with hands-on experience in Generative AI, multi-agent systems, and RAG pipelines, having built production-level projects using LangGraph, Groq, and Streamlit.

Technical Skills

Programming Languages: Python (Intermediate), SQL, C
Generative AI & Agentic Systems: LangGraph, Multi-Agent Systems, RAG, MCP, Prompt Engineering, LLM Integration
Machine Learning: Supervised Learning, Feature Engineering, Demand Forecasting, Model Evaluation
Optimization & Backend: MILP, PuLP, FastAPI, PostgreSQL, REST APIs
Data Analytics & BI: Power BI (DAX, Power Query), Excel, pandas, NumPy, Statistical Hypothesis Testing
Development Tools: Streamlit, Git, GitHub, ChromaDB, Vector Databases

Experience

Data Science Intern Dec 2024 – Jan 2025
NLC India Limited *Neyveli, Tamil Nadu*

- Built data processing workflows and performed transformations using Power Query and DAX on 1500+ transaction records.
- Worked with procurement datasets by applying cleaning, merging, and transformation techniques.
- Created interactive Power BI dashboards with optimized data models for business reporting.

Projects

Agentic Inventory Intelligence System | *Python, LangGraph, FastAPI, PostgreSQL, PuLP, LLMs*

- Built a **LangGraph multi-agent system** for demand forecasting, inventory, and supplier performance analysis across regions.
- Developed a **PuLP-based MILP engine** for cost-effective and constraint-aware procurement optimization under business constraints.
- Created a **natural-language parser** to convert user requests into structured constraints and objective optimization weights.
- Designed **supplier capacity checks** with strict-capacity and proportional-sizing optimization modes for feasibility assessment.
- Integrated **FastAPI and PostgreSQL** for decision persistence, constraint validation, and explainable procurement analytics.

AI Travel Planning System | *Python, LangGraph, Groq API, PostgreSQL, MCP, Streamlit*

- Built a **multi-agent AI system** using **LangGraph**, with specialized agents for flights, hotels, and itinerary planning through a shared state graph.
- Integrated **MCP servers** to provide agents with a standardized interface for calling external tools independently of their implementation.
- Implemented **PostgreSQL checkpointing** to persist user preferences and previous travel plans across multiple sessions.
- Designed **state-based agent orchestration** where each agent processes and passes shared context to downstream agents.
- Built a **Streamlit interface** for natural-language trip planning with MCP-based tool calls under budget and destination constraints.

Agentic RAG System | *Python, Generative AI, LLM APIs, Vector Database*

- Built an **RAG-based chatbot** that retrieves relevant information from uploaded documents to generate context-grounded answers.
- Converted documents into **vector embeddings** and stored them in **ChromaDB** for semantic similarity search.
- Used **LangGraph** to intelligently route queries between the RAG pipeline and direct LLM responses based on user intent.
- Integrated **Groq LLMs** with prompt engineering to generate accurate, context-grounded responses while reducing hallucinations.
- Developed a **Streamlit interface** supporting document upload, chatbot interaction, model selection, and source-cited responses.

Education

B.Tech Artificial Intelligence and Data Science (Expected 2027) **CGPA: 7.89**
Rajalakshmi Engineering College, Chennai

Jawahar Higher Secondary School [CBSE] **79.4%**
Year of Passing: 2023

Extracurricular / Certifications

HackerRank SQL (Basic) Certification - Certificate

Deloitte Forage Virtual Internship – Data Analytics - Simulation